

ASSIGNMENT 1

Mathematics for Computing

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Outline

- ① Examples of Set-Builder Form
- ② Set Theory Terminologies
- ③ Set Operations
- ④ Hasse Diagrams
- ⑤ General Inclusion–Exclusion Principle Proof

Set-Builder Form to Roster Form

1

$$\begin{aligned} \{x \mid x = n^2, n \in \mathbb{Z}, 0 \leq n \leq 6\} \\ = \{0, 1, 4, 9, 16, 25, 36\} \end{aligned}$$

2

$$\begin{aligned} \{x \mid x = 2^n, n \in \mathbb{N}\} \\ = \{2, 4, 8, 16, 32, \dots\} \end{aligned}$$

3

$$\begin{aligned} \{x \mid x \mid 12, x \in \mathbb{N}\} \\ = \{1, 2, 3, 4, 6, 12\} \end{aligned}$$

Set-Builder Form to Roster Form

4

$$\begin{aligned}\{x \mid 3 \mid x, -6 \leq x \leq 6, x \in \mathbb{Z}\} \\ = \{-6, -3, 0, 3, 6\}\end{aligned}$$

5

$$\begin{aligned}\{x \mid x \equiv 1 \pmod{2}, -5 \leq x \leq 5, x \in \mathbb{Z}\} \\ = \{-5, -3, -1, 1, 3, 5\}\end{aligned}$$

6

$$\begin{aligned}\{(x, y) \mid y = x^2, -2 \leq x \leq 2, x, y \in \mathbb{Z}\} \\ = \{(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)\}\end{aligned}$$

7

$$\begin{aligned}\{(x, y) \mid x = y, 1 \leq x \leq 3, x \in \mathbb{N}\} \\ = \{(1, 1), (2, 2), (3, 3)\}\end{aligned}$$

8

$$\left\{x \mid x = \frac{1}{n}, n \in \mathbb{N}\right\}$$

Set-Builder Form to Roster Form

9

$$\{(x, y) \mid x \in \{1, 2\}, y \in \{a, b\}\}$$

$$= \{(1, a), (1, b), (2, a), (2, b)\}$$

10

$$\{(x, y) \mid x = n^2, y = -5, n \in \mathbb{N}\}$$

$$= \{(1, -5), (4, -5), (9, -5), \dots\}$$

Cardinality

The cardinality of a set A , denoted by $|A|$, is the number of distinct elements in the set.

Example

Let

$$F = \{\text{CPU, RAM, Disk, Bandwidth, Latency}\}$$

Then,

$$|F| = 5$$

This represents the number of features used in a system-performance prediction model.

Finite Set

A set A is called finite if there exists a natural number $n \in \mathbb{N}$ such that

$$|A| = n.$$

In other words, a set is finite if it contains a finite number of elements.

Example

A feature vector contains four attributes:

$$F = \{f_1, f_2, f_3, f_4\}$$

Hence, F is finite.

Infinite Set

A set A is called infinite if it is not finite.

Equivalently, A is infinite if there does not exist any natural number $n \in \mathbb{N}$ such that

$$|A| = n.$$

Example

The set of all integer-valued model parameters:

$$W = \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

Hence, W is infinite.

Null Set

A set A is called a null set if it has no elements.
Its cardinality is zero.

$$A = \emptyset = \{\}$$

Example

Suppose a database query searches for models trained in the year 2100.

$$M = \{x \mid x \text{ was trained in 2100}\}$$

If no such model exists,

$$M = \emptyset.$$

Equal Set

Two sets A and B are called equal sets if they contain exactly the same elements, irrespective of their order.

$$A = B \iff \forall x (x \in A \iff x \in B)$$

Example

Let

$$A = \{\text{Python, C++}, \text{Java}\}$$

and

$$B = \{\text{Java, Python, C++}\}.$$

Then,

Equivalent Set

Two sets A and B are called equivalent sets if they have the same cardinality.

$$A \sim B \iff |A| = |B|$$

Example

Let

$$A = \{\text{SVM, Decision Tree, Naive Bayes}\}$$

and

$$B = \{\text{CNN, RNN, Transformer}\}.$$

Since

$$|A| = |B| = 3,$$

Subset and Proper Subset

A set A is a subset of a set B if every element of A is also an element of B .

$$A \subseteq B \iff \forall x (x \in A \Rightarrow x \in B)$$

Example

Let

$$F = \{\text{CPU, RAM, Disk, Bandwidth}\}$$

and

$$G = \{\text{CPU, RAM}\}.$$

Then,

$$G \subseteq F.$$

Super Set

A set A is a superset of a set B if every element of B is also an element of A .

$$A \supseteq B \iff \forall x (x \in B \Rightarrow x \in A)$$

Example

Let

$$F = \{\text{CPU, RAM, Disk, Bandwidth}\}$$

and

$$G = \{\text{CPU, RAM}\}.$$

Then,

$$F \supseteq G.$$

Power Set

The power set of a set A is the set of all subsets of A .

It is denoted by

$$\mathcal{P}(A)$$

or

$$2^A.$$

$$\mathcal{P}(A) = \{X \mid X \subseteq A\}$$

Example

For feature selection,

$$F = \{f_1, f_2, f_3\}.$$

Then

Singleton Set

A set A is called a singleton set if it contains only one element.

$$A = \{a\}$$

Therefore,

$$a \in A$$

and

$$|A| = 1.$$

Example

A graph with one vertex:

$$V = \{v_1\}$$

Universal Set

A set U is called a universal set if every set being considered is a subset of U .
Every element of any set A belongs to the universal set U .

$$\forall A, \quad A \subseteq U$$

Example

Let U be the set of all records in a training database:

$$U = \{r_1, r_2, \dots, r_n\}.$$

Any subset of records is a subset of U .

Countable Set

A countable set is a set whose elements can be listed or counted one by one, even if the set has infinitely many elements.

A set A is countable if

$$|A| \leq |\mathbb{N}|.$$

Example

The set of all finite binary strings:

$$B = \{\epsilon, 0, 1, 00, 01, 10, 11, 000, \dots\}$$

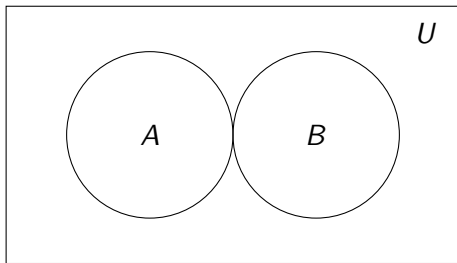
is countable because the strings can be ordered by length and lexicographic order.

Union

The union of two sets is the set containing all elements that belong to either set.

The union of sets A and B is

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$



Example – Union

Machine Learning Datasets

Let

$$A = \{d_1, d_2, d_3, d_4\}$$

be the set of training samples collected from Source 1.

Let

$$B = \{d_3, d_4, d_5, d_6\}$$

be the set of training samples collected from Source 2.

Then,

$$A \cup B = \{d_1, d_2, d_3, d_4, d_5, d_6\}.$$

Application:

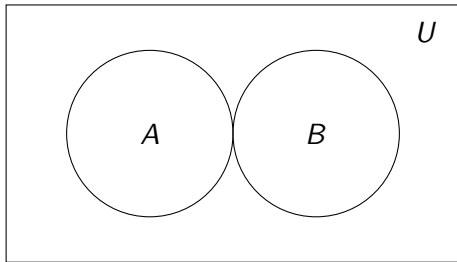
Combining datasets from multiple sources to create a larger training dataset.

Intersection

The intersection of two sets is the set containing only the elements that are common to both sets.

The intersection of sets A and B is

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$



Example – Intersection

Feature Selection

Let

$$A = \{f_1, f_2, f_3, f_4, f_5\}$$

be features extracted from Dataset 1.

Let

$$B = \{f_2, f_4, f_6, f_7\}$$

be features selected by another algorithm.

Then,

$$A \cap B = \{f_2, f_4\}.$$

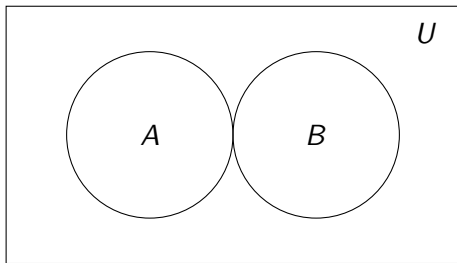
Application:

Finding common important features selected by different feature-selection methods.

Set Difference

The set difference of two sets is the set of elements that belong to the first set but not to the second set.

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}.$$



Example – Set Difference

Data Preprocessing

Let

$$A = \{r_1, r_2, r_3, r_4, r_5\}$$

be all records in a database.

Let

$$B = \{r_2, r_5\}$$

be records removed because they contain missing values.

Then,

$$A - B = \{r_1, r_3, r_4\}.$$

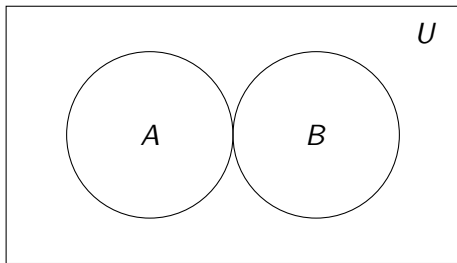
Application:

Obtaining valid training records after removing corrupted data.

Symmetric Difference

The symmetric difference of two sets is the set of elements that belong to either set but not to both.

$$A \triangle B = \{x \mid (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$$



Example – Symmetric Difference

Algorithm Comparison

Let

$$A = \{\text{Algm1}, \text{Algm2}, \text{Algm3}\}$$

be algorithms implemented in System A.

Let

$$B = \{\text{Algm2}, \text{Algm4}, \text{Algm5}\}$$

be algorithms implemented in System B.

Then,

$$A \triangle B = \{\text{Algm1}, \text{Algm3}, \text{Algm4}, \text{Algm5}\}.$$

Application:

Finding algorithms that exist in only one software system.

Complement

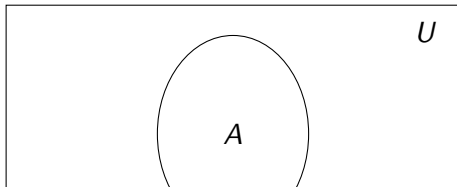
The complement of a set A is the set of all elements in the universal set that are not in A . It is denoted by

$$A^c$$

or

$$A'.$$

$$A^c = \{x \mid x \in U \text{ and } x \notin A\}.$$



Example – Complement

Dataset Classes

Universal set:

$$U = \{\text{Class1}, \text{Class2}, \text{Class3}, \text{Class4}, \text{Class5}\}$$

Classes selected for training:

$$A = \{\text{Class1}, \text{Class3}\}$$

Then,

$$A^c = \{\text{Class2}, \text{Class4}, \text{Class5}\}.$$

These are the classes not selected.

Cartesian Product

The Cartesian product of two sets is the set of all ordered pairs formed by taking the first element from the first set and the second element from the second set.

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Example – Computer Networks

Let

$$A = \{192.168.1.1, 192.168.1.2\}$$

be a set of IP addresses.

Let

$$B = \{80, 443\}$$

be a set of ports.

Then,

Example 1

Hasse diagram of $(A, \leq)$, where

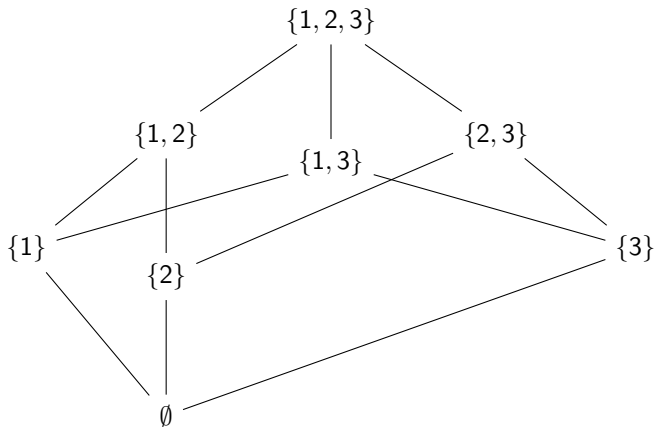
$$A = \{1, 2, 3, 4, 5\}.$$



Example 2

Hasse diagram of $(\mathcal{P}(A), \subseteq)$, where

$$A = \{1, 2, 3\}.$$



General Inclusion–Exclusion Principle

Theorem (General Inclusion–Exclusion Principle)

Let

$$A_1, A_2, \dots, A_n$$

be finite sets. Then

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| &= \sum |A_i| - \sum |A_i \cap A_j| \\ &\quad + \sum |A_i \cap A_j \cap A_k| - \dots \\ &\quad + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|. \end{aligned}$$

Proof.

Suppose an element x belongs to exactly r of the sets.

In the first sum, the element x is counted r times:

$$\binom{r}{1}.$$

General Inclusion–Exclusion Principle – Proof

In the second sum, x is counted once for every pair of the r sets containing x .

Therefore, it is subtracted:

$$-\binom{r}{2}.$$

In the third sum, it is counted once for every triple:

$$+\binom{r}{3}.$$

Thus, the counting continues as

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1} \binom{r}{r}.$$

This process continues alternately by subtracting four-set intersections, adding five-set intersections, and so on.

General Inclusion–Exclusion Principle – Proof

Using the binomial identity,

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1} \binom{r}{r} = 1.$$

Therefore, each element is counted exactly once.

Hence,

$$\begin{aligned} |A_1 \cup A_2 \cup \cdots \cup A_n| &= \sum |A_i| - \sum |A_i \cap A_j| \\ &\quad + \sum |A_i \cap A_j \cap A_k| - \cdots \\ &\quad + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n|. \end{aligned}$$

Hence proved.